KARATINA UNIVERSITY

1. Distinguish between the following terms as used in statistics:

   i) A parameter and a statistic. [2 marks]

   ii) Type I error and Type II error. [2 marks]

- ✓ A parameter is a measurable characteristic pertaining to the whole population.
- ✓ A statistic is a characteristic measured from the sample
- ✓ Point estimator- a single number which is computed from a sample.
- ✓ An interval estimator is an interval obtained from a sample
- ✓ Type I error – rejecting H_0 when it should be accepted
- ✓ Type ii error- accepting H_0 when it should be rejected

2. Consider a computer system with Poisson job-arrival stream at an average of 2 per minute. Determine the probability that in any one-minute interval there will be more than 3 arrivals. [3 marks]

Solution

$$P(X > 3) = 1 - P(X \leq 3) = 1 - \{P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)\}$$

$$= 1 - \left\{\frac{e^{-2}2^0}{0!} + \frac{e^{-2}2^1}{1!} + \frac{e^{-2}2^2}{2!} + \frac{e^{-2}2^3}{3!}\right\} = 1 - 0.8571 = 0.1429$$

3. The masses of packages from a particular machine are normally distributed with mean of 200g and standard deviation of 2 g. Find the probability that a randomly selected package weighs less than 197 g. [3 marks]

Solution

Let x weight of a given package

$x \sim N(200,4)$

$$P(x < 197) = P\left(\frac{x-\mu}{2} < \frac{197-200}{2}\right)$$

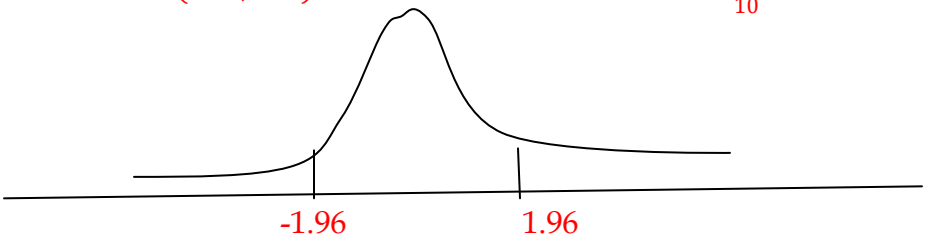
$$P(Z < -1.50) = 1 - \emptyset(1.5) = 1 - 0.9332 = 0.0668$$

4. A measure of a random variable is taken at random and found to be 172. Test at 5% level of significance whether the population mean could be 150 given that the standard deviation is 100. [3 marks]

$H_0: \mu = 150$

$H_1: \mu \neq 150$

$X \sim N(150,100)$ The test statistics $Z = \frac{172-150}{10} = 2.2$



Since z I s greater than 1.96 which is the critical value so we reject the null hypothesis

We conclude the significant evidence in 5% to suggest that the mean is not 150

At a supermarket, 80% of customers pay using credit cards. Find the probability that in a randomly selected sample of 20 customers exactly 2 pay using the card [3 marks]

Solution

i) Define x

$x = number\ of\ customers\ paying\ using\ the\ card$

$x \sim b(20,0.8)$

$f(x) = \binom{20}{x} 0.8^x \times 0.2^{20-x}$

Exactly 2 pay using the card

$f(20) = P(X = 2)$

$P(X = 2) = f(2) = \binom{20}{2} 0.8^2 \times 0.2^{18} = 3.1877 \times 10^{-11}$

5. A small village has 6 households as depicted in table below

Household	A	B	C	D	E	F
Household size	8	6	3	5	4	4

Draw all possible simple random samples of size 2 (without replacement). [2 marks]

$ab, ac, ad, ae, af, bc, bd, be, bf, cd, ce, cf, de, df, ef$

15 samples are drawn

a) A manufacturer of safety pins knows that 5 % of his product is defective. If he sells safety pins in boxes of 100 and guarantees that not more than 5 pins will be defective, what is the approximate probability that a box will fail to meet the guaranteed quality? [4 marks]

Solution

A box fails to meet the guaranteed quality if it contains more than 5 pins that is
$P(X > 5)$ mean= $\lambda = np = 100 \times 0.05 = 5$

Therefore $P(X > 5) = 1 - P(X \leq 5)$

$$1 - \sum_{x=0}^{5} \frac{e^{-5}5^x}{x!}$$

$$= 1 - \left[\frac{e^{-5}5^0}{0!} + \frac{e^{-5}5^1}{1!} + \frac{e^{-5}5^2}{2!} + \frac{e^{-5}5^3}{3!} + \frac{e^{-5}5^4}{4!} + \frac{e^{-5}5^5}{5!}\right]$$

$$= 1 - [0.0067 + 0.337 + 0.0842 + 0.140 + 0.1755 + 0.1755]$$

$$1 - 0.6156 = 0.3844$$

b) A sample of 11 circuits from a large normal population has a mean resistance of 2.20 ohms. We know from past testing that the population standard deviation is 0.35 ohms. Determine a 95% confidence interval for the true mean resistance of the population. [3 marks]

$$\mu = 2.20 \pm 1.96 \times \frac{0.35}{\sqrt{11}} = 2.20 \pm 0.2068$$

The interval is $1.9932 \leq \mu \leq 2.4068$

c) A random sample of 50 account balances for a firm extracted from a large number of debtors account gave a mean debit account balance 6, 432 with standard deviation 1650. What sample size would be required if the population estimates is to be within Ksh 300 and at 98% confidence level. [3 marks]

Solution

$\alpha = 2\% = 0.02$ $Z_\alpha = 2.33$

$\mu = \bar{x} \pm 300$

$$\Rightarrow Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} = 300 \Rightarrow n = \left(\frac{2.33 \times 16.50}{200}\right)^2 = 12.815^2$$

$$= 164.22$$

Hence sample size n≅ 164

A random sample of eight drivers insured with a company and having similar auto insurance policies was selected. The following table lists their driving experiences (in years) and monthly auto insurance premiums.

Driving Experience (years)	5	2	12	9	15	6	25	16
Monthly Auto Insurance Premium	64	87	50	71	44	56	42	60

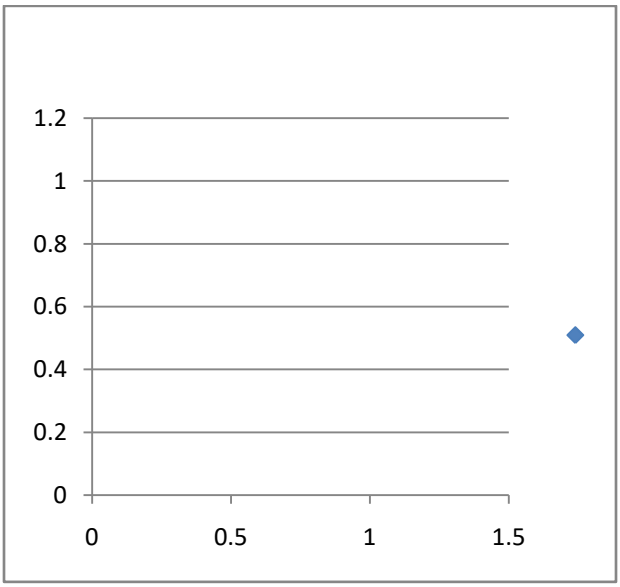
a) .Does the insurance premium depend on the driving experience or does the driving experience depend on the insurance premium? Justify your answer [1 marks]

b) Plot the scatter diagram and comment on the correlation between the two variables. [2 marks]

c) Find the least squares regression line by choosing appropriate dependent and independent variables based on your answer in part (a). [4 marks]

a) Based on theory and intuition, it's expected the insurance premium to depend on driving experience. Consequently, the insurance premium is a dependent variable and driving experience is an independent variable in the regression model.

b) Scatter graph



Comment – shows negative correlation or negative linear relationship

c) Least square regression line

Experience	Premium	xy	x^2	y^2
5	64	320	25	4096
2	87	174	4	7569
12	50	600	144	2500
9	71	639	81	5041
15	44	660	225	1936
6	56	336	36	3136
25	42	1050	625	1764
16	60	960	256	3600
90	**474**	**4739**	**1396**	**29642**

$$b_1 = \frac{8 \times\times 4739 - 90 \times 474}{8 \times\times 1396 - 90^2} = -1.5476$$

$$b_o = \frac{476}{8} - (-1.5476) \times \frac{90}{8} = 76.66$$

Thus the regression line is Type equation here.

$$\hat{y} = 76.66 - 1.545x$$

REVISION QUESTIONS

1. The probability that a pig dies of food poisoning is 0.001. Determine the probability that out of 1000 pigs.

   i) Expected number suffering from the disease. [1 mark]

   $$E(x) = np = 1000 \times 0.001 = 1$$

   ii) Standard deviation. [1mark]

   $$Var(x) = npq = 1000 \times 0.001 \times 0.999 = 0.999, \quad sd = 0.9995$$

   iii) Exactly two will die of food poisoning. [1 mark]

$$P(x=2)=\frac{e^{-1}1^2}{2}=0.1839$$

2. The marks of 500 of students in a test are normally distributed with mean 45 and standard deviation 20. Given that the pass mark is 41 estimate the number of students that passed. [2 marks]

Solution

Let x be the number of marks obtained by students

$x \sim N(45,20^2)$

$P(x \geq 41) = \quad P(z \geq \frac{41-45}{20})$

$P(Z \geq -0.2) = \emptyset(0.2) = 0.5793$

Number of students who passed $= 0.5793 \times 500 = 289$

3. The mass in grams of packets of biscuits of a particular brand is thought to follow the normal distribution. A random sample

12.3, 11.8, 12.4, 11.9, 12.8 *and* 12.1

is taken.

i) Test the hypothesis that the random sample came from a normal population with mean 12.0 and a known standard deviation of 0.4. (Use α=5%) [3 marks]

ii) Repeat Q (i) above assuming the standard deviation is not known. [2 marks]

Solution

a) $\bar{X}=\frac{12.3+15.8+12.4+11.9+11.8+12.1}{6}=12.22$

i) σ is known use Ƶ

$H_o: \mu = 12.0$

$H_0: \mu \neq 12.0$

$\text{Ƶ}=\frac{\bar{X}-\mu}{\frac{\sigma}{\sqrt{n}}}=\frac{12.22-12.0}{\frac{0.4}{\sqrt{6}}}=1.35$

But $\text{Ƶ}_{\frac{\alpha}{2}}=\text{Ƶ}_{0.025}=1.96$

$\Rightarrow$calculated $\text{Ƶ}=1.35 < table\ \text{Ƶ}_{\frac{\alpha}{2}}=1.96$ hence accept H_o

ii) σ Not known. Hence use t-test

$$S = \sqrt{\frac{\sum(x-\bar{X})^2}{n-1}} = \sqrt{\frac{\sum x^2 - n\bar{X}}{n-1}} = \sqrt{\frac{896.15-6\times149.246}{5}} = \sqrt{0.1337} \cong 0.366$$

Therefore $t = \frac{\bar{X}-\mu}{\frac{s}{\sqrt{n}}} = \frac{12.22-12}{\frac{0.366}{\sqrt{6}}} = 1.47$

But $t_{\frac{\alpha}{2}} = t_{0.025} with\ (6-1) = 5$ degrees of freedom $= 2.5706$

Therefore calculated $t = 1.47 < table\ t_{\frac{\alpha}{2}} = 2.5706$ hence accept H_o

4. Consider a study involving Metal Tags on Penguins. The researcher performs a test for the difference in the proportion of penguins who survive over a 10-year period, between penguins tagged with metal tags and those tagged with electronic tags. The objective is in testing whether the type of tag has an effect on penguin survival rate, using a chi-square test. In the study, 33 of the 167 metal-tagged penguins survived while 68 of the 189 electronic-tagged penguins survived.

   a) Create a two-way table from the information given. {2 marks}

   b) State the null and alternative hypotheses. [1 marks]

   c) Give a table with the expected counts for each of the four categories. [2 marks]

   d) Calculate the chi-square test statistic. [3 marks]

   e) Determine the p-value and state the conclusion. [1 marks]

Solution

(a) The two-way table of penguin survival vs type tag is shown:

	Metal	Electronic	Total
Survived	33	68	101
Died	134	121	255
Total	167	189	356

(b) The hypothesis are

H_0: Type of tag is not related to survival

H_a: Type of tag is related to survival

(c) The table below shows the expected counts, obtained for each cell by multiplying the row total by the column total and dividing by $n = 356$.

	Metal	Electronic
Survived	47.4	53.6
Died	119.6	135.4

d) We calculate the chi-square test statistic

$$\begin{aligned}\chi^2 &= \frac{(33-47.4)^2}{47.4} + \frac{(68-53.6)^2}{53.6} + \frac{(134-119.6)^2}{119.6} + \frac{(121-135.4)^2}{135.4} \\ &= 4.37 + 3.87 + 1.73 + 1.53 \\ &= 11.5\end{aligned}$$

(e) We compare our test statistic of 11.5 from part (c) to a chi-square with 1 degree of freedom to get a p-value of 0.0007. There is strong evidence that the type of tag and survival rate of the penguins are related.

Consider a case of a company that markets and repairs small computers. To study the relationship between the length of a service call and the number of electronic components in the computer that must be repaired or replaced, a sample of records on service calls was taken. The data consist of the length of services calls in minutes (the response variable) and the number of components repaired (the predictor variable).

Row	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Minutes	23	29	49	64	74	87	96	97	109	119	149	145	154	166
Units	1	2	3	4	4	5	6	6	7	8	9	9	10	10

a) Determine the regression line for

Suppose that the management expected the increase service time for each additional unit to be repaired to be 12 minutes. Do the data support this conjecture? (Hint: standard error of $\beta_1 = 0.505$, $\left(t_{n-2}, \frac{\alpha}{2}\right) = 2.18$. [13 marks]

i	Y_i	X_i	$(y_i - \bar{y})^2$	$(x_i - \bar{X})^2$	$(y_i - \bar{y})(x_i - \bar{X})$
1	23	1	5507.76	25	371.07
2	29	2	4653.19	16	272.86
3	49	3	2324.62	9	144.64
4	64	4	1103.19	4	66.43
5	74	4	538.9	4	46.43
6	87	5	104.33	1	10.21
7	96	6	1.47	0	0
8	97	6	0.05	0	0
9	109	7	138.9	1	11.79
10	119	8	474.62	4	43.57
11	149	9	2681.76	9	155.36
12	145	9	2283.47	9	143.14
13	154	10	3224.62	16	227.14
14	166	10	4731.47	16	275.14
Totals	1361	84	27768.36	114	1768.0

$\bar{Y} = \frac{\sum y_i}{n} = \frac{1361}{14} = 97.21$

$\bar{X} = \frac{\sum x_i}{n} = \frac{84}{14} = 6$

$cov(Y, X) = \frac{\sum(Y_i - \bar{Y})(X_i - \bar{X})}{\sqrt{\sum(Y_i - \bar{Y})^2}\sqrt{\sum(X_i - \bar{X})^2}} = \frac{1768}{\sqrt{27768.36 \times 114}} = 0.996$

Regression equation $Y = b_0 + b_1 X$

Where $b_1 = \frac{\sum(Y_i - \bar{Y})(X_i - \bar{X})}{\sum(X_i - \bar{X})^2} = \frac{1768}{114} = 15.509$

$b_0 = \bar{Y} - b_1\bar{X} = 97.21 - 15.509(6) = 4.162$

Equation minutes= $4.162 + 15.509$ units

$H_0: B_1 = 12\ Vs\ \ H_1 = B_1: 12$

$t_1 = \frac{\beta_1 - 12}{Se(B_1)} = \frac{15.509 - 12}{0.505} = 6.948$

$(t_{\left(n-2, \frac{\alpha}{2}\right)} = 2.18$ With 12 degrees of freedom

Highly significant because $t_1 > t\ tabulated$. Reject H_0 and accept H_1.

The management estimate of the increase in time for each additional component to be repaired is not supported by the data.

a) Batteries of a radio have a mean life of 160 hours with a standard deviation of 30 hours. Assuming normal distribution. Calculate the percentage of batteries which have a life of between 150 hours and 180 hours. [4 marks]

Solution

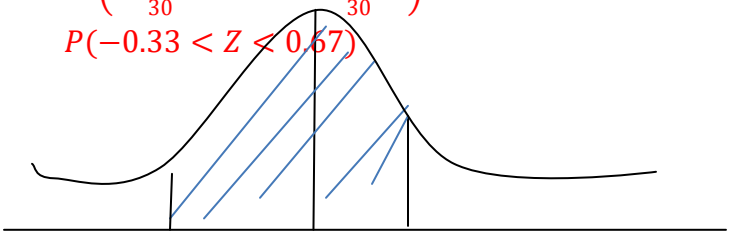
i) Let x be the number of hours of life batteries

$x \sim N(160, 30^2)$

$P(150 < x < 180)$

$P\left(\frac{150-160}{30} < Z < \frac{180-160}{30}\right)$

$P(-0.33 < Z < 0.67)$



-0.33 0.67

$P(-0.33 < Z < 0.67) = \emptyset(0.67) + \emptyset(0.6293) - 1 = 0.3779 = 37.79\%$

ii) Assume that lifetime of cells are independent.

$P(X \geq 135)$

$$P\left(Z \geq \frac{135 - 160}{30}\right) = P(Z \geq 0.8333)$$

a) The marks of 500 of students in a test are normally distributed with mean 45 and standard deviation 20. Given that the pass mark is 41 estimate the number of students that passed. [4 marks]

Solution

Let x be the number of marks obtained by students

$x \sim N(45, 20^2)$

i) $P(x \geq 41) =$ $P(z \geq \frac{41-45}{20})$

$P(Z \geq -0.2) = \emptyset(0.2) = 0.5793$

Number of students who passed $= 0.5793 \times 500 = 289$

ii) $P(Z \geq a) = 0.05$,

$a = \emptyset^{-1}(0.95)$ $a = 1.65$

$Z = \frac{x-\mu}{\sigma} \quad \Rightarrow \quad a = \frac{x-45}{20}$

$x = 1.65 \times 20 + 45 = 78$

b) A sample of 200 people with a particular disease was selected. Out of these, 100 were given a drug and the others were not given any drug. The results are as follows;

Number of people			
	Drug	No drug	Total
Cured	65	55	120
Not cured	35	45	80
Total	100	100	200

Test whether the drug is effective or not at 5% level of significance. [5 marks]

Solution

Use χ^2 test. Let H_o be the drug is not effective in curing the disease

Observed	Expected	$(0-E)^2$	$\frac{(0-E)^2}{E}$
65	60	25	0.417
35	40	25	0.625
55	60	25	0.417
45	40	25	0.625
			$\frac{\sum(0-E)^2}{E} = 2.084$

Then $\chi^2 = \frac{\sum(0-E)^2}{E} = 2.084$ degrees of freedom $v = (r-1)(c-1) = (2-1)(2-1) = 1$ from the tables $v = 1, \chi^2_{0.05} = 3.841$

Calculated $\chi^2 <$ tabulated $\chi^2_{0.05}$

Conclusion: accept H_o drug is not effective in curing the disease

KARATINA UNIVERSITY

COURSE CODE: STA 205

COURSE TITLE: PROBABILITY AND STATISTICS

QUESTION ONE (30 MARKS)

3. Distinguish between the following terms as used in statistics:

iii) A parameter and a statistic. [2 marks]

iv) Type I error and Type II error. [2 marks]

✓ A parameter is a measurable characteristic pertaining to the whole population.

- ✓ A statistic is a characteristic measured from the sample
- ✓ Point estimator- a single number which is computed from a sample.
- ✓ An interval estimator is an interval obtained from a sample
- ✓ Type I error – rejecting H_0 when it should be accepted
- ✓ Type ii error- accepting H_0 when it should be rejected

4. Explain the hypothesis testing procedure. [4 marks]

Solution (deny 2 marks if explanation is omitted)

- ✓ **Formulate the a hypothesis**
- ✓ **Set up a suitable significance level**
- ✓ **determination of a suitable test statistic**
- ✓ **Determine the critical region**
- ✓ **Doing Computations**
- ✓ **Making decisions**

5. The continuous variable X has a probability function

$$f(x) = \begin{cases} \frac{1}{4}(2x+3), 0 \le x \le 1 \\ 0, elsewhere \end{cases}$$

i) Show that $f(x)$ is a probability density function. [3 marks]

ii) Find $P(0 \le x \le \frac{1}{2})$. [3 marks]

Solution

$$\int_x f(x)\,dx = 1 \Rightarrow \int_0^1 \frac{1}{4}(2x+3)dx = \frac{1}{4}\left[x^2 + 3x\right]_{x=0}^{1} = 1$$

i) We need to show

Hence f (x) is a p. d. f.

ii) $\Pr(0 \le X \le \frac{1}{2}) = \int_{x=0}^{1/2} \frac{1}{4}(2x+3)\,dx = \frac{7}{16}$

6. A discrete random variable Y has a probability mass function given by the table below;

Y	0	1	2	3
P(Y=y)	A	3a	3a	a

i) Find the value of the constant a [2 marks]

ii) Compute $P(1 \le Y < 3)$ [2 marks]

Solution

i) For a pmf $\sum_{\forall i} p_i = 1, P(1 \le Y \le 4) = a(1 + 3 + 3 +) = 1$ Hence $a = \frac{1}{8}$

ii) thus $P(1 \le Y < 3) = \frac{1}{8}(1+3) = \frac{1}{2}$

7. Consider a computer system with Poisson job-arrival stream at an average of 2 per minute. Determine the probability that in any one-minute interval there will be more than 3 arrivals. [5 marks]

   Solution

   $$P(X > 3) = 1 - P(X \leq 3) = 1 - \{P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)\}$$

   $$= 1 - \left\{\frac{e^{-2}2^0}{0!} + \frac{e^{-2}2^1}{1!} + \frac{e^{-2}2^2}{2!} + \frac{e^{-2}2^3}{3!}\right\} = 1 - 0.8571 = 0.1429$$

8. The masses of some components for computer hardwares are normally distributed with mean of 200g and standard deviation of 2 g. Find the probability that a randomly selected package weighs less than 197 g. [3 marks]

   Solution

   Let x weight of a given package

   $x \sim N(200,4)$

   $$P(x < 197) = P(\frac{x-\mu}{2} < \frac{197-200}{2})$$

   $$P(Z < -1.50) = 1 - \emptyset(1.5) = 1 - 0.9332 = 0.0668$$

9. The marks of 500 of students in a data base management test are normally distributed with mean 45 and standard deviation 20. Given that the pass mark is 41 estimate the number of students that passed. [4 marks]

Solution

Let x be the number of marks obtained by students

$x \sim N(45,20^2)$

$$P(x \geq 41) = P(z \geq \frac{41-45}{20})$$

$$P(Z \geq -0.2) = \emptyset(0.2) = 0.5793$$

Number of students who passed $= 0.5793 \times 500 = 289$

QUESTION TWO (20 Marks)

a) Telephone calls arrive at a switchboard at the rate of 50 per hour. Find the probabilities of 0 or1 or 2 calls arriving in any 5 minute period.

[5 marks]

Solution

The average rate of call per 5 minute period $= \frac{50}{12} = 4.17 calls$.

If the random variable X is "the number of calls in any 5 minute period",

then X is Po (4.17).

$$\Pr(X = x) = f(x) = \frac{e^{-4.17}(4.17)^x}{x!}, \quad x = 0, 1, 2, 3, \cdots$$
$$= 0, \; elsewhere$$

$$\Pr(X = 0) = f(0) = \frac{e^{-4.17}(4.17)^0}{0!} = 0.02$$

$$\Pr(X = 1) = f(1) = \frac{e^{-4.17}(4.17)^1}{1!} = 0.06$$

$$\Pr(X = 2) = f(2) = \frac{e^{-4.17}(4.17)^2}{2!} = 0.13$$

so $p(0 \; or \; 1 \; or \; 2) = 0.02 + 0.06 + 0.13 = 0.21$

b) A telephone operator receives 6 calls on average between $10 - 11 \; am$ on a weekday. Find the probability that he receives 2 or more calls during this time on Monday. [5 marks]

Assume x forms a Poisson distribution

Let x be the number of calls received between 10-11 am, $\lambda = 6$

$$P(X \geq 2) = 1 - P(X \leq 1)$$

$$P(X \leq 1) = P(X = 0) + P(X = 1)$$

$$P(X = x) = f(x) = \frac{e^{-\lambda}\lambda^x}{x!} \quad P(X = 0) = \frac{e^{-6}6^0}{0!} = 0.025$$

$$P(X = 1) = \frac{e^{-6}6^1}{1!} = 0.0149$$

$$P(X \geq 2) = 1 - 0.0174 = 0.9826$$

c) On average, 20% of bolts produced by a machine in a factory are faulty. A sample of 10 bolts is selected at random. Find the probability;

i) Exactly 2 bolts are faulty. [3 marks]

ii) More than 3 bolts are faulty [4 marks]

iii) At least 2 bolts are faulty. [3 marks]

Solution

ii) Define x

$x = number\ of\ faulty\ bolts$

$x \sim b(10, 0.2)$

$f(x) = \binom{10}{x} 0.8^x \times 0.2^{20-x}$

Exactly 2 pay using the card

$f(20) = P(X = 2)$

$P(X = 2) = f(2) = \binom{20}{2} 0.8^2 \times 0.2^{18} = 3.1877 \times 10^{-11}$

iii) More than 3 $P(X > 30) = 1 - P(X \leq 3)$

$P(X \leq 3) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$

$P(X = 0) = 20c_0 0.8^0 (0.2)^{20} = 1.0486 \times 10^{-14}$

$P(X = 1) = 20c_1 0.8^1 (0.2)^{19} = 8.3886 \times 10^{-13}$

$P(X = 2) = 3.1877 \times 10^{-11}$

$P(X \leq 3) = 20c_3 0.8^3 (0.2)^{17} = 7.6504 \times 10^{-10}$

$1 - 7.9777 \times 10^{-10} = 0.99999 \cong 1$

iv) At least 2 pay using card

$P(X \geq 2) = 1 - P(X \leq 1)$

$P(X \leq 1) = P(X = 0) + P(X = 1)$

$P(X = 0) = 20c_0 0.8^0 (0.2)^{20} = 1.0486 \times 10^{-14}$

$P(X = 1) = 20c_1 0.8^1 (0.2)^{19} = 8.3886 \times 10^{-13}$

$P(X \geq 2) = 1 - (0.8443 \times 10^{-14}) = 1$

QUESTION THREE (20 Marks)

A computer Scientist is analyzing the performance of a computer algorithm based on the size of the input data. He collects data on the size of the input (in kilobytes) and the corresponding execution time (in milliseconds) for various inputs and recorded in the table below.

Input size(in kilobytes)	1	2	3	4	5	6	7	8
Execution time (in milliseconds)	12.5	15	17.5	20	22.5	25	27.5	30

d) Plot the scatter diagram and comment on the correlation between the two variables and comment on the correlation.

[4 marks]

e) Find the least squares regression line of y on x [8 marks]

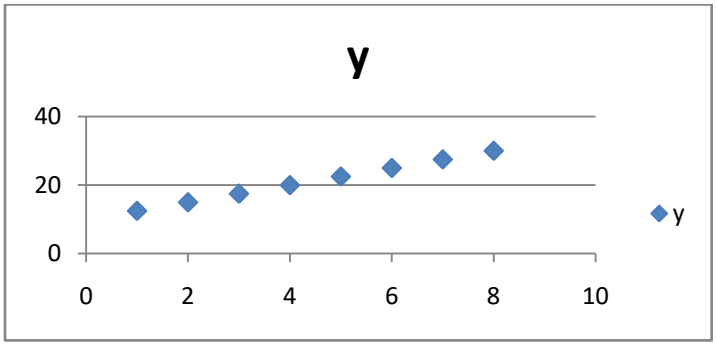
f) Predict the execution time for 10 kilobytes. [1 mark]

g) Given that the execution time is 26 milliseconds, find the size of the input.[2 marks]

h) Calculate the coefficient of determination r^2 and comment on its value. [5 marks]

Solution

d) Scatter graph



The graph shows perfect positive linear relationship

e) Least square regression line

x	**y**	**x^2**	**xy**	y^2
1	12.5	1	12.5	156.25
2	15	4	30	225
3	17.5	9	52.5	306.25
4	20	16	80	400
5	22.5	25	112.5	506.25
6	25	36	150	625
7	27.5	49	192.5	756.25
8	30	64	240	900
36	**170**	**204**	**870**	3875

$$b_1 = \frac{8 \times 870 - 36 \times 170}{8 \times 3875 - 36^2} = 2.5$$

$$b_o = \frac{170}{8} - (2.5) \times \frac{36}{8} = 10$$

Thus the regression line is Type equation here.

$$\hat{y} = 10 + 2.5x$$

f) The value of a = 10 gives the value of ŷ for x = 0; that is, it gives the execution time when there is no imput
The value of b gives the change in ŷ due to a change of one unit in x. Thus, b = 2.5 indicates that, on average, for every extra size of input, the the execution time increases by 2.5.

Using the estimated regression line, we find the predicted value of y for x = 10 is $\hat{y} = 10 + 2.5 \times 10 = 35\ kilobytes$

g) $\hat{y} = 10 + 2.5x = 26 \quad x = 6.4\ milliseconds$
h) Thus, we expect the monthly auto insurance premium of a driver with 10 years of driving experience to be $61.18.
i) $r^2 = 1$ (show working)

The value of $r^2 = 1$ states that 100% of the total variation in execution time is explained by the input ssize.

QUESTION FOUR (20 Marks)

a) In a computer science company, the time taken by employees to resolve a certain type of technical issue follows a normal distribution with a mean of 45 minutes and a standard deviation of 8 minutes.
   i) Determine the probability that a randomly selected employee will take between 35 and 50 minutes to resolve the technical issue? [4 marks]
   ii) If the company wants to ensure that at least 80% of technical issues are resolved within a certain time frame, what should be the maximum time allowed for resolving an issue? [4 marks]

Solution:

Given parameters: Mean (μ) = 45 minutes and (σ) = 8 minutes

i) $P(35 \leq X \leq 50) = P\left(-\frac{10}{8} \leq Z \leq \frac{5}{8}\right) = P(-1.25 \leq Z \leq 0.625)$

$= 0.735 - 0.1056 = 0.6310$

ii) To ensure that at least 80% of technical issues are resolved within a certain time frame, we need to find the time value corresponding to the z-score that encapsulates 80% of the data.

$P\left(z \leq \frac{x-45}{8}\right) = 0.8$.

$z = \sigma x - \mu, \quad x = 45 + 8 \times 0.845 = 51.76$

So, the maximum time allowed for resolving an issue to ensure that at least 80% of technical issues are resolved within that time frame is approximately 51.76 minutes.

b) A group of computer scientists wants to determine whether there is a significant difference in the programming language preferences among software developers based on their expertise level. They surveyed 500 software developers and categorized them into three groups based on their expertise level: Beginner, Intermediate, and Advanced. The survey results are summarized in the table below:

programming language	Beginner	Intermediate	advanced
Python	80	90	100
Java	60	70	50
C++	40	60	70
Java Scriptx	50	80	60

To determine whether there is a significant association between expertise level and programming language preferences among software developers.

i) State the null and alternative hypotheses. [2 marks]

ii) Give a table with the expected counts for each of the four categories. [6 marks]

iii) Calculate the chi-square test statistic. [3 marks]

iv) Interpret the statistic obtained in (iii) at a significance level of 0.05 and state the conclusion. [2 marks]

Solution

Solution

Step 1: Formulate Hypotheses

Null Hypothesis (H0): There is no significant association between expertise level and programming language preferences among software developers.

Alternative Hypothesis (H1): There is a significant association between expertise level and programming language preferences among software developers.

Step 2: Choose Significance Level

α = 0.05

Step 3: Set Up Contingency Table

- Expected frequency for Python and Beginner: $\frac{(250\times230)}{810} \approx 70.82$
- Expected frequency for Python and Intermediate: $\frac{(250\times300)}{810} \approx 92.59$
- Expected frequency for Python and Advanced: $\frac{(250\times280)}{810} \approx 86.59$
- Expected frequency for Java and Beginner: $\frac{(200\times230)}{810} \approx 56.44$
- Expected frequency for Java and Intermediate: $\frac{(200\times300)}{810} \approx 74.07$
- Expected frequency for Java and Advanced: $\frac{(200\times280)}{810} \approx 69.48$
- Expected frequency for C++ and Beginner: $\frac{(170\times230)}{810} \approx 48.19$
- Expected frequency for C++ and Intermediate: $\frac{(170\times300)}{810} \approx 62.96$
- Expected frequency for C++ and Advanced: $\frac{(170\times280)}{810} \approx 58.85$
- Expected frequency for JavaScript and Beginner: $\frac{(190\times230)}{810} \approx 54.55$
- Expected frequency for JavaScript and Intermediate: $\frac{(190\times300)}{810} \approx 71.30$
- Expected frequency for JavaScript and Advanced: $\frac{(190\times280)}{810} \approx 66.15$

Now, we'll compute the chi-square test statistic using the formula:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Where O_i is the observed frequency and E_i is the expected frequency.

Let's compute the chi-square test statistic.

To compute the chi-square test statistic, we'll use the formula:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Where O_i is the observed frequency and E_i is the expected frequency.

We'll sum the contributions from each cell:

$$\chi^2 = \frac{(60 - 70.82)^2}{70.82} + \frac{(90 - 92.59)^2}{92.59} + \frac{(100 - 86.59)^2}{86.59} + \frac{(80 - 56.44)^2}{56.44} + \frac{(70 - 74.07)^2}{74.07} + \frac{(50 - 69.48)^2}{69.48} + \frac{(40 - 48.19)^2}{48.19} + \frac{(60 - 62.96)^2}{62.96} + \frac{(70 - 58.85)^2}{58.85} + \frac{(50 - 54.55)^2}{54.55} + \frac{(80 - 71.30)^2}{71.30} + \frac{(60 - 66.15)^2}{66.15}$$

Now, we'll compute this sum.

$$\chi^2 \approx 5.098 + 0.036 + 1.749 + 9.790 + 0.206 + 3.181 + 1.360 + 0.154 + 1.634 + 0.279 + 1$$

$$\chi^2 \approx 24.148$$

Now, we compare this value to the critical value from the chi-square distribution table with 6 degrees of freedom and a significance level of 0.05. If the calculated chi-square value exceeds the critical value, we reject the null hypothesis.

Let's compare this value to the critical value.

The calculated chi-square value is $\chi^2 \approx 24.148$.

Now, we compare this value to the critical value from the chi-square distribution table with 6 degrees of freedom and a significance level of 0.05. If the calculated chi-square value exceeds the critical value, we reject the null hypothesis.
Let's compare this value to the critical value.
The calculated chi-square value is $\chi 2 \approx 24.148$.
Now, we compare this value to the critical value from the chi-square distribution table with 6 degrees of freedom and a significance level of 0.05.
From the chi-square distribution table, the critical value for 6 degrees of freedom and a significance level of 0.05 is approximately 12.592.
Since the calculated chi-square value (24.148) exceeds the critical value (12.592), we reject the null hypothesis.
Step 8: Make a Decision
Since the calculated chi-square value exceeds the critical value, we reject the null hypothesis.
Step 9: Conclusion
At a significance level of 0.05, there is sufficient evidence to conclude that there is a significant association between expertise level and programming language preferences among software developers.

QUESTION FIVE (20 Marks)

10. Explain the meaning of the following terms as used in statistics:
    v) A random variable . [1 mark]
    vi) Discrete density function. [2 marks]

Solution

i) **A random variable** is a rule for assigning a particular number to a particular experimental outcome.

ii) Any function f with domain on the real line and counter domain or range [0, 1] is defined to be a discrete density function if for some countable set $x_1, x_2, \cdots, x_n$;

$f(x_i) > 0, \ for \ i = 1,2,\cdots,n, \qquad f(x) = 0, \quad for \ x \neq x_i, \quad i = 1,2,\cdots,n,$

$\sum_x f(x_i) = 1,$ where the summation is over the points $x_1, x_2, \cdots, x_n$.

c) The mass in grams of packets of biscuits of a particular brand is thought to follow the normal distribution. A random sample

12.3, 11.8, 12.4, 11.9, 12.8 *and* 12.1

is taken. Test the hypothesis that the random sample came from a normal population with mean 12.0 and a known standard deviation of 0.4. (Use α=5%)

[6 marks]

Solution

b) $\bar{X} = \frac{12.3+15.8+12.4+11.9+11.8+12.1}{6} = 12.22$

iii) σ is known use Ƶ

$H_o: \mu = 12.0$

$H_0: \mu \neq 12.0$

$Ƶ = \frac{\bar{X}-\mu}{\frac{\sigma}{\sqrt{n}}} = \frac{12.22-12.0}{\frac{0.4}{\sqrt{6}}} = 1.35$

But $Ƶ_{\frac{\alpha}{2}} = Ƶ_{0.025} = 1.96$

$\Longrightarrow$calculated $Ƶ = 1.35 < table \ Ƶ_{\frac{\alpha}{2}} = 1.96$ hence accept H_o

vii)A computer scientist wants to evaluate whether a new algorithm they have developed improves the performance of a certain task compared to the existing algorithm. The task involves sorting a large dataset of integers. The scientist collects data on the time taken (in milliseconds) to sort the dataset using both the new and existing algorithms. The collected data is as follows:

viii)

	Sample size	Mean	Standard deviation
New Algorithm	25	52	8
Existing Algorithm	25	58	10

Assuming the time taken to sort the dataset follows a normal distribution for both algorithms, conduct a hypothesis test at a significance level of 0.05 to determine

whether there is sufficient evidence to conclude that the new algorithm improves the performance compared to the existing algorithm

[8 marks]

Solution:

Step 1: Formulate Hypotheses
Null Hypothesis (H0): The new algorithm does not improve the performance compared to the existing algorithm. (μ_new = μ_existing)
Alternative Hypothesis (H1): The new algorithm improves the performance compared to the existing algorithm. (μ_new < μ_existing)
Step 2: Choose Significance Level
α = 0.05
Step 3: Compute Test Statistic

$$t = \frac{(\bar{x}_{new} - \bar{x}_{existing})}{\sqrt{\frac{s^2_{new}}{n_{new}} + \frac{s^2_{existing}}{n_{existing}}}}$$

Where:

- $\bar{x}_{new}$ = Sample mean for the new algorithm
- $\bar{x}_{existing}$ = Sample mean for the existing algorithm
- s_{new} = Sample standard deviation for the new algorithm
- $s_{existing}$ = Sample standard deviation for the existing algorithm
- n_{new} = Sample size for the new algorithm
- $n_{existing}$ = Sample size for the existing algorithm

Given: $\bar{x}_{new}$=52 milliseconds, s_{new}=8 milliseconds,
n_{new}=25, $\bar{x}_{existing}$=58 milliseconds, $s_{existing}$=10 milliseconds, $n_{existing}$=25

$$t = \frac{(52-58)}{\sqrt{\frac{8^2}{25} + \frac{10^2}{25}}}$$
$$t = \frac{-6}{\sqrt{\frac{64}{25} + \frac{100}{25}}}$$
$$t = \frac{-6}{\sqrt{\frac{164}{25}}}$$
$$t = \frac{-6}{\sqrt{6.56}}$$
$$t \approx \frac{-6}{2.56}$$
$$t \approx -2.34$$

Step 4: Determine Critical Value

Since this is a one-tailed test (alternative hypothesis is directional), we look up the critical value from the t-distribution table for a significance level of 0.05 and degrees of freedom $df=nnew+nexisting-2=25+25-2=48$.
For $df=48$ and a one-tailed test with $a=0.05$, the critical value is approximately -1.677.
Step 5: Make a Decision
Since the calculated test statistic $t=-2.34$ is less than the critical value of -1.677, we reject the null hypothesis.
Step 6: Conclusion

At a significance level of 0.05, there is sufficient evidence to conclude that the new algorithm improves the performance compared to the existing algorithm.

11. Distinguish between the following terms; cluster sampling and stratified sampling. [2 marks]

Solution

Cluster sampling is accomplished by dividing the population into groups- - usually geographically. These groups are called clusters or blocks. The clusters are randomly selected, and each element in the selected clusters is used.

Stratified sampling also divides the population into groups called strata. However, this time it is by some characteristic, not geographically. For instance, the population might be separated into males and females. A sample is taken from each of these strata using either random, systematic, or convenience sampling.

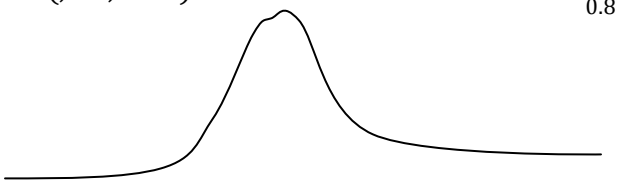
12. A company in the information technology sector collects a large number of website response times before the implementation of a new website. Test a hypothesis that the response time of the large number of responses is 3.0 seconds if a sample of 100 responses is used such that the sample mean is 3.6 seconds and the standard deviation is 0.8 seconds. [4 marks]

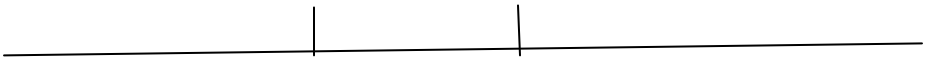
solution

$H_0: \mu = 3.0$

$H_1: \mu \neq 3.0$

$X \sim N(,3.6, 0.64)$ The test statistics $\mathrm{Z} = \frac{3.6-3}{0.8} = 0.75$





Since z is less than 1.96 which is the critical value so we do not reject the null hypothesis

We conclude there is significant evidence in 5% to suggest that the mean is 3.0